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Cost Estimation

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1. Cost associated with Constructed Facilities

- **Capital cost related to the initial establishment**
  - Land acquisition, including assembly, holding and improvement
  - Planning and feasibility studies
  - Architectural and engineering design
  - Construction, including materials, equipment and labor
  - Field supervision of construction
  - Construction financing
  - Insurance and taxes during construction
  - Owner's general office overhead
  - Equipment and furnishings not included in construction
  - Inspection and testing
- **Operation and maintenance cost**
  - Land rent, if applicable
  - Operating staff
  - Labor and material for maintenance and repairs
  - Periodic renovations
  - Insurance and taxes
  - Financing costs
  - Utilities
  - Owner's other expenses

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1. Cost associated with Constructed Facilities

- The resources demands for three types of major energy projects investigated during the energy crisis in the 1970's are shown in the table. These projects are: (1) an oil shale project with a capacity of 50,000 barrels of oil product per day; (2) a coal gasification project that makes gas with a heating value of 320 billions of British thermal units per day, or equivalent to about 50,000 barrels of oil product per day; and (3) a tar sand project with a capacity of 150,000 barrels of oil product per day.

TABLE 5-1 Resource Requirements of Some Major Energy Projects

	Oil shale (50,000 barrels/day)	Coal gasification (320 billions BTU/day)	Tar Sands (150,000 barrels/day)
Cost ($ billion)	2.5	4	8 to 10
Basic design (Thousands of hours)	80	200	100
Detailed engineering (Millions of hours)	3 to 4	4 to 5	6 to 8
Construction (Millions of hours)	20	30	40
Materials ($ billion)	1	2	2.5

Source: Exxon Research and Engineering Company, Florham Park, NJ

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2. Approaches to Cost Estimation

1. Production function

- In microeconomics, the relationship between the output of a process and the necessary resources is referred to as the production function.
- In construction, the production function may be expressed by the relationship between the volume of construction and a factor of production such as labor or capital.

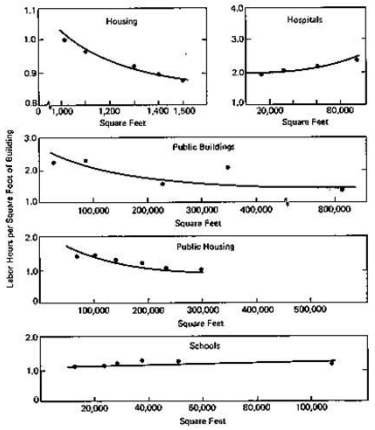


Figure 3-3: Illustrative Relationships between Building Size and Input Labor by Types of Building
(Reprinted with permission from P.J. Cassimatis, *Economics of the Construction Industry*, The National Industry Conference Board, SEB, No. 111, 1969, p.53)

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2. Approaches to Cost Estimation

2. Unit costs for bill of quantities

- Unit method
  e.g. cost of hospital = 300 beds @ $1M/bed
- Floor area method
  Cost per m2 of floor area
  Arcadis's Quarter Construction Cost Review

Building Type	HK$/m2 CFA	Outline Specification
tic		
nts, high rise, public authority standard	11,200 - 13,400	Apartment units with fit-out, based on Hong Kong Housing Auth Non-standard Cruciform Block design (Rental Housing)
nts, high rise, average standard	24,100 - 27,900	Apartment units with fit-out, including air-conditioning, kitchen cabinets and home appliances, but excluding decorative light fit and loose furniture
nts, high rise, high end	31,200 - 36,200	Apartment units with good quality fit-out, including air- conditi kitchen cabinets and home appliances, but excluding decorative fittings and loose furniture
houses, average standard	33,200 - 38,300	Houses with fit-out, including air-conditioning, kitchen cabinets home appliances, but excluding decorative light fittings, loose fu garden and parking
f houses, high end	48,300 up	Houses with good quality fit-out, including air-conditioning, kitch cabinets and home appliances, but excluding decorative light fit loose furniture, garden and parking
Commercial		
high rise offices, average standard	23,900 - 27,200	RC structure, curtain wall/window wall, including public area fit- tenant area with screeded floor, painted wall and ceiling
offices, prestige quality	28,700 - 32,900	RC structure, curtain wall, including public area fit-out, tenant ar raised floor, painted wall and false ceiling
wn shopping centre, average standard	23,600 - 27,700	Including public area fit-out and M&E, but excluding shop fit-out
ills, high end	30,700 - 36,400	
ial		
erated factories, low rise, light dustry	18,300 - 23,000	RC structure, including small office with simple fit-out and M&E, excluding a/c

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2. Approaches to Cost Estimation

3. Allocation of joint costs



LABOR, MATERIAL, CONSTRUCTION EQUIPMENT, CONSTRUCTION SUPERVISION, GENERAL OFFICE OVERHEAD

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3. Types of construction cost estimates

1. **Design Estimates.**

For the owner or its designated design professionals, the types of cost estimates encountered run parallel with the planning and design as follows:

1. Screening estimates (or order of magnitude estimates)
2. Preliminary estimates (or conceptual estimates)
3. Detailed estimates (or definitive estimates)
4. Engineer's estimates based on plans and specifications

For each of these different estimates, the amount of design information available typically increases.

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3. Types of construction cost estimates

2. Bid Estimates.

For the contractor, a bid estimate submitted to the owner either for competitive bidding or negotiation consists of direct construction cost including field supervision, plus a markup to cover general overhead and profits. The direct cost of construction for bid estimates is usually derived from a combination of the following approaches.

1. Subcontractor quotations
2. Quantity takeoffs
3. Construction procedures.

3. Control Estimates. For monitoring the project during construction, a control estimate is derived from available information to establish:

1. Budget estimate for financing
2. Budgeted cost after contracting but prior to construction
3. Estimated cost to completion during the progress of construction.

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4. Prediction of price movements

Tender price index (TPI)

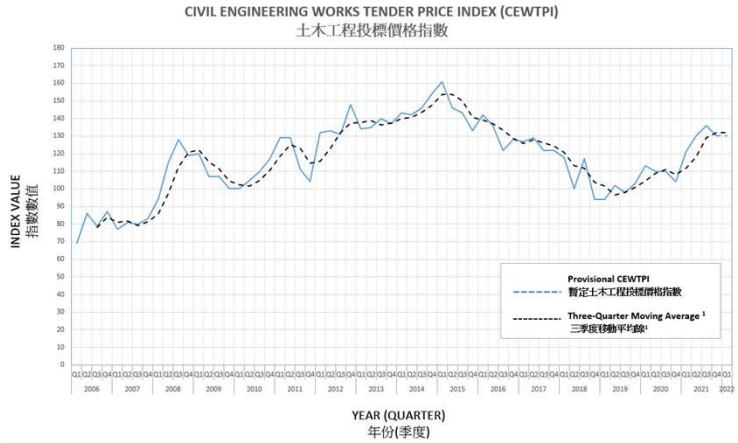
- Tender price = construction costs + markup
  TPI – derived by averaging the prices of contracts awarded in each quarter
- Usage
  Depict historical market conditions (establish trends)
  Predict future tender prices

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5. Civil Engineering Works Tender Price Index (CEWTPI)

CEDD Civil Engineering and Development Department
The Government of the Hong Kong Special Administrative Region



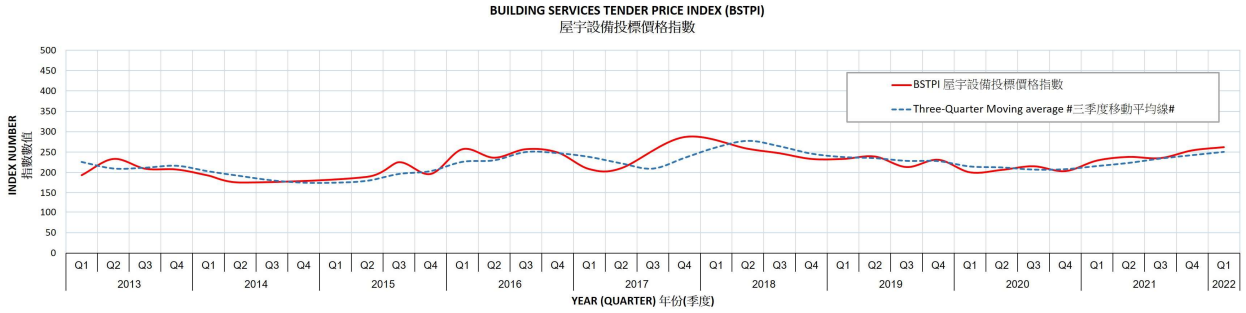
- CEWTPI is compiled in good faith to provide (i) an indication of the level of tender prices for civil engineering works (excluding architectural, E&M installations, ground investigation, soft landscaping, other ancillary type of works and term contract works) undertaken by the Civil Engineering and Development Department, Highways Department, Water Supplies Department and Drainage Services Department and (ii) an aid to adjust civil engineering works cost data for estimating purposes.

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6. Building Services Tender Price Index (BSTPI)

Architectural Services Department
The Government of the Hong Kong Special Administrative Region



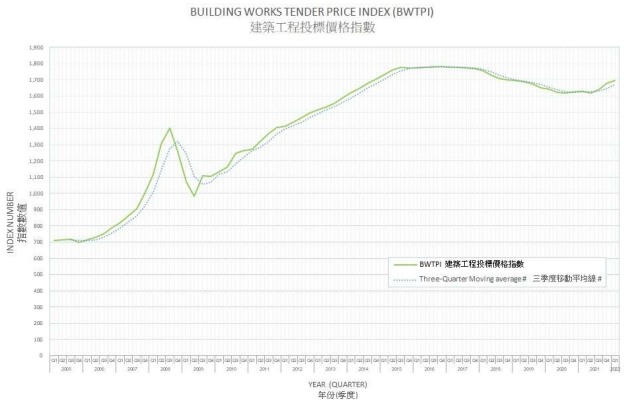
BSTPI is compiled by ArchSD in good faith to provide (i) an indication of the level of tender prices for building services installation in new building works under Capital Works Programme and (ii) an aid to adjust building services installation cost data for estimating purposes.

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6. Building Works Tender Price Index (BWTPI)

Architectural Services Department
The Government of the Hong Kong Special Administrative Region



Building Works Tender Price Index (BWTPI)

BWTPI is compiled by ArchSD in good faith to provide (i) an indication of the level of tender prices for **new building works** (excluding building services works) under Capital Works Programme and (ii) an aid to adjust building cost data for estimating purposes.

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7. Arcadis

Building & Building Services Price Index
Mainland China



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8. Building cost index (BCI)

Building cost

- Building cost = cost of all resources (i.e. labour, plant & material) for a project which is paid by the contractor
- It does not reflect market condition (i.e. profit)
- BCI is constantly upward, while TPI can fluctuate

Usage

- Calculate fluctuation (if any)
- Bring a tender figure (without markup) for a previous project up to current prices

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9. Census

香港特別行政區 政府統計處
Census and Statistics Department
Hong Kong Special Administrative Region

公營建築工程的工資及材料成本指數（2021 年 4 月 = 100）#
INDEX NUMBERS OF THE COSTS OF LABOUR AND MATERIALS USED IN PUBLIC SECTOR CONSTRUCTION PROJECTS (APRIL 2021 = 100) #

2022年2月
FEBRUARY 2022

項目	Item	指數數值 Index Numbers
工資	Labour	
綜合工資（土木工程合約）	Composite labour wages for civil engineering contracts	102.5
綜合工資（建築合約）	Composite labour wages for building contracts	100.3
特選材料	Selected materials	
碎石	Aggregates	95.4
瀝青	Bitumen	113.3
混凝土磚	Concrete blocks	122.1
柴油	Diesel fuel	115.4
玻璃	Glass	112.3
釉面牆壁瓷磚	Glazed ceramic wall tiles	103.3
木工	Woodwork	99.3
均質地磚	Homogeneous floor tiles	111.0
鍍鋅軟鋼	Galvanised mild steel	122.2
金屬模板	Metal formwork	118.0
馬賽克	Mosaic tiles	111.8
油漆 *	Paint *	105.8
波特蘭水泥（普通）	Portland cement (ordinary)	117.6
砂 ^	Sand ^	102.6
鋼筋	Steel reinforcement	112.6
木模板	Timber formwork	108.3
硬膠喉管	uPVC pipes	104.7
鍍水喉管	GMS pipes	115.5
銅喉	Copper pipes	111.7
紅筒	Epoxy coated cast iron pipes	110.6

註釋：

工資及材料成本指數是用於計算公營建築工程因應成本變動而調整主要承建商所收款項的增減額，因此，工資指數只涵蓋公營建築工程及反映公營建築工程主要承建商所承受工資成本的變動。

* 數字包括低揮發性有機化合物油漆。

^ 數字不包括海砂及機製砂。

Notes:

The index numbers of the costs of labour and materials are compiled for working out the increase/decrease of payment to the main contractors of public sector construction projects to adjust for the change in costs. The costs of labour indices therefore only cover public sector construction projects and reflect the changes in labour costs borne by main contractors of public sector construction projects.

* Figure includes low volatile organic compound paint.

^ Figure does not include marine sand and manufactured sand.

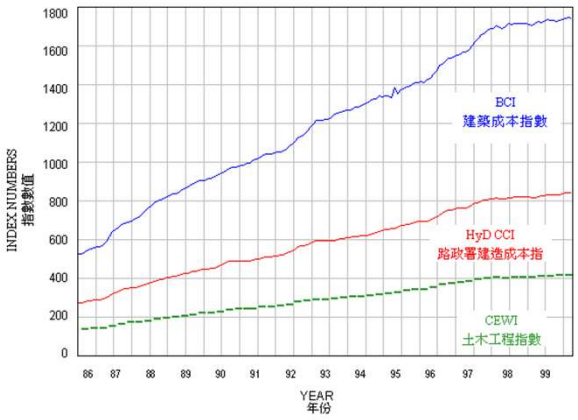
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Architectural Services Department Building Cost Index (BCI)

Highways Department Construction Cost Index (HyD CCI)

Civil Engineering Works Index (CEWI)



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